

Kinematic Synchronization Buffer — System Analysis

1. Introduction

Many automated production and logistics systems share a common structural challenge: a stream of discrete objects arrives at irregular intervals from an upstream source and must be handed off to a downstream process that expects them at precise, fixed intervals. The upstream and downstream are mechanically decoupled — they run at independently set speeds — and a buffer element in between is responsible for absorbing the timing mismatch and delivering each object into the correct position at the correct moment.

This document analyses one realization of that problem: the **Kinematic Synchronization Buffer (KSB)**. The KSB is a servo-driven conveyor segment that receives objects one at a time from a conditioning stage and synchronizes each one to a target slot on a continuously moving downstream carrier. Synchronization is achieved by executing a jerk-limited motion profile over the object's transit through the buffer, adjusting its position so that it arrives at the handoff point at the right time to enter its assigned slot.

The system operates under a kinematic constraint: the motion profile must respect bounds on velocity, acceleration, and jerk, and each object must reach its target within the time window defined by the downstream carrier's slot timing. When the upstream arrival rate is close to the downstream slot rate, the kinematic demands on the buffer are modest. The interesting regime — and the focus of this analysis — is when the rates differ slightly, forcing the buffer to occasionally skip a downstream slot and assign an object to the next available one. These **skip events** are the primary driver of kinematic pressure on the buffer's edge segments and the primary source of the failure modes characterized in the failure-modes document.

The treatment in this document is structural. The buffer is decomposed into independently servo-controlled segments, and the per-segment formalism developed in §2 — the transition $\zeta_{i,k}$ a segment must execute, the free window $W_{i,k}$ available to execute it, and the minimum time $T_{\min}(\zeta_{i,k})$ required under the segment's bounds — is the language in which the rest of the work is conducted. §3 reads the segment's job as a control problem, noting that the synchronization deficit admits two equivalent descriptions — a time error and a state error — that carry the same information but lend themselves to different uses.

The core questions this document addresses are:

- What are the structural invariants of the system — quantities that depend only on the rate ratio and not on arrival variability?

- How does upstream arrival variability change the character of skip events and their consequences for the buffer's synchronization demands?
- What failure-mode quantity best captures where the buffer's capacity is being exceeded, and how does it behave as variability and load change?

The analysis proceeds from first principles, defining the per-segment quantities and the kinematic regime in which each policy operates, and arriving at the two failure modes — timing collision and kinematic shortfall — that the failure-modes document then characterizes empirically. The treatment is intentionally general: while the KSB is one physical instantiation, the same mathematical structure appears in any system where a stochastic arrival stream must be synchronized to a deterministic slot sequence under kinematic constraints — a pattern that recurs in packaging, logistics sortation, assembly line feeding, and discrete-event manufacturing more broadly.

2. The Buffer

2.1 Internal Structure

The KSB is a servo-driven buffer segment of length L^B that sits between the conditioning stage and the downstream carrier. Its job is to transport each input from the entry boundary to the handoff boundary while executing a jerk-limited motion profile that synchronizes the input to its assigned downstream slot.

The buffer is internally divided into N^B segments. Segment k , for $k \in \{0, \dots, N^B - 1\}$, has length L_k^B , with $\sum_{k=0}^{N^B-1} L_k^B = L^B$. In buffer-local coordinates with the entry boundary at zero, the upstream boundary of segment k is at $p_k = \sum_{j=0}^{k-1} L_j^B$ (so $p_0 = 0$) and its downstream boundary is at $p_k + L_k^B$. The downstream boundary of segment $N^B - 1$ is at L^B .

Each segment is independently servo-controlled and operates at its own velocity. The straddling constraint of the system formalization applies at every internal boundary: when an input straddles the boundary between segments k and $k + 1$ — leading edge past $p_{k+1} = p_k + L_k^B$, trailing edge before it — both segments must run at matched velocity. The same constraint applies at the buffer's two outer boundaries, where the adjacent stages (conditioning conveyor, downstream carrier) likewise lock to the buffer's edge segments during straddling.

There is a mechanical lower bound $L_{\min}^B$ on segment length, reflecting limits on how short an independently-controlled servo segment can physically be built. This bound is structural, not derived from synchronization requirements, and is taken as a given throughout this document.

2.2 Per-Segment Quantities

Each input is modeled as a point of mass located at its leading edge, with the trailing edge at $p_i(t) - l$ where $p_i(t)$ is the leading-edge position of input i at time t and l is the (uniform) input

length. The buffer transports inputs in one direction only; the point-mass model loses nothing relative to a centroid or extended-body formulation and simplifies all downstream definitions.

Crossing events. Two events define segment k 's control window for input i :

$$t_{i,k}^{\text{in}} := \text{time } t \text{ such that } p_i(t) = p_k$$

— input i 's leading edge reaches segment k 's upstream boundary; segment k begins to bear input i . And:

$$t_{i,k}^{\text{out}} := \text{time } t \text{ such that } p_i(t) = p_k + L_k^B + l$$

— input i 's trailing edge clears segment k 's downstream boundary; segment k is fully clear of input i .

Segment states at crossing events. During the intervals when input i is being borne by segment k , segment k 's velocity is locked by the straddling constraint to match input i 's velocity. At each of the two crossing events, segment k 's state is recorded:

$$\xi_{i,k}^{\text{in}} := (v_{i,k}^{\text{in}}, a_{i,k}^{\text{in}}) \quad \xi_{i,k}^{\text{out}} := (v_{i,k}^{\text{out}}, a_{i,k}^{\text{out}})$$

For internal segments, $\xi_{i,k}^{\text{in}}$ is inherited from segment $k - 1$'s handoff and $\xi_{i,k}^{\text{out}}$ is delivered to segment $k + 1$ at handoff. The two outer edges of the buffer are exceptions: $\xi_{i,0}^{\text{in}}$ is inherited from the conditioning stage rather than from an internal segment, and $\xi_{i,N^B-1}^{\text{out}}$ is delivered to the downstream carrier rather than to an internal segment. At the entry edge the inherited state is the conditioned arrival — velocity v_c and the stretched pitch the conditioning stage produces; at the exit edge the delivered state is pinned to what the slot demands. Asymmetric behavior at these edge boundaries is taken up in the failure-modes document.

Free windows. Between consecutive inputs i and $i + 1$, segment k has an interval during which no input is on it — the *free window*. It opens at $t_{i,k}^{\text{out}}$ (input i leaves) and closes at $t_{i+1,k}^{\text{in}}$ (input $i + 1$ arrives):

$$W_{i,k} := t_{i+1,k}^{\text{in}} - t_{i,k}^{\text{out}}.$$

During this window, segment k is uncoupled from any input and is free to swing its velocity from $v_{i,k}^{\text{out}}$ to $v_{i+1,k}^{\text{in}}$ under the synchronization policy π^S .

Transitions. The transition segment k must execute over the free window between inputs i and $i + 1$ is the pair of states at the two endpoints:

$$\zeta_{i,k} := (\xi_{i,k}^{\text{out}}, \xi_{i+1,k}^{\text{in}}).$$

Minimum-time function. Under π^S 's kinematic bounds (acceleration, jerk; see §2.4), the minimum duration required to execute the transition $\zeta_{i,k}$ is

$T_{\min}(\zeta_{i,k}) :=$ Ruckig duration for $\zeta_{i,k}$ under π^S bounds, with no minimum-duration constraint.

This is an operational definition: $T_{\min}$ is what Ruckig returns when called with boundary conditions $\zeta_{i,k}$ and π^S 's kinematic bounds. The synchronization is feasible on segment k between inputs i and $i + 1$ exactly when

$$W_{i,k} \geq T_{\min}(\zeta_{i,k}).$$

The signed quantity $M_{i,k} := W_{i,k} - T_{\min}(\zeta_{i,k})$ is called the **kinematic margin** on segment k between inputs i and $i + 1$ — positive when the segment has temporal headroom on the transition, negative when it has a deficit. The feasibility test rewrites as $M_{i,k} \geq 0$, and the magnitude $|M_{i,k}|$ measures by how much the segment is in surplus or deficit. The kinematic margin is the central diagnostic quantity in the failure-modes characterization and in the subsystem rationales that follow.

Three regimes of the margin. The sign of $W_{i,k}$ partitions the margin into three regimes, distinguishable without evaluating $T_{\min}$:

- $W_{i,k} < 0$ — a **timing collision**. Input $i + 1$ reaches segment k 's upstream boundary before input i has cleared its downstream boundary. Two inputs are simultaneously on the segment, which can command only one velocity; no π^S trajectory exists at all. The margin is necessarily negative.
- $0 \leq W_{i,k} < T_{\min}(\zeta_{i,k})$ — a **kinematic shortfall**. The segment is clear long enough for a transition to begin, but the window is too short for π^S to complete it under its bounds. A trajectory exists but does not fit. The margin is negative.
- $W_{i,k} \geq T_{\min}(\zeta_{i,k})$ — **feasible**. The transition completes within the window and the margin is non-negative.

The two failure regimes are the two failure modes of the buffer, characterized in the failure-modes document. Operationally the partition is convenient: the timing collision is detected by the sign of $W_{i,k}$ alone, before any solver is invoked, and only the intermediate band requires evaluating $T_{\min}$ to classify.

The structural skip mechanism that drives the distribution of $W_{i,k}$ across batch position is set by the load ratio $\rho = r_u/r_d$ and the resulting mean skip interval $Q = 1/(1 - \rho)$, both defined in the system formalization. ρ and Q are recalled here as needed without re-introduction.

2.3 Physical Assumptions

Two physical assumptions underlie the formalization above and are made explicit here so the rest of the document can rest on them without restating them at each use.

Zero-slip at input/segment contact. While an input is being borne by a segment, the input moves at the segment's velocity with no slippage. This is the basis for the straddling constraint

and for the recording of segment states at the crossing events $t_{i,k}^{\text{in}}$ and $t_{i,k}^{\text{out}}$. The zero-slip assumption also fixes the kinematic bounds available to the per-input control policy π^I : the acceleration and jerk a segment may impose on a borne input cannot exceed the contact-friction limit, beyond which the assumption fails and the input slides. See §2.4.

Perfect synchronization under $W \geq T_{\min}$. The synchronization policy π^S achieves the demanded transition $\zeta_{i,k}$ exactly when the free window is sufficient, $W_{i,k} \geq T_{\min}(\zeta_{i,k})$. There is no partial credit and no degradation policy: the transition is either completed within the window, in which case π^S delivers the demanded $\xi_{i+1,k}^{\text{in}}$ at $t_{i+1,k}^{\text{in}}$, or it cannot be completed, in which case the synchronization is infeasible and the kinematic-shortfall failure mode is recorded. The assumption is that the policy itself does not introduce error within its feasibility region — the only question is whether the region contains the demanded transition. This makes feasibility a binary property at the policy level and frames the analysis as a characterization of where the feasibility boundary sits.

2.4 Kinematic Bounds: π^I vs π^S

The two policies π^I (per-input control) and π^S (segment synchronization) operate under different sets of kinematic bounds, because the physical regime they operate in is different.

π^I bounds: zero-slip at contact. While a segment bears an input, the input must not slip. The maximum acceleration the segment can impose without slip is $a_{\max}^I = \mu g$, where μ is the static friction coefficient between input and segment surface and g is gravitational acceleration. The jerk bound $j_{\max}^I$ is set by servo capability under the no-slip regime (the friction limit applies to acceleration, but the rate of change of acceleration is constrained by what the servo can deliver while remaining within the acceleration budget). These bounds apply to every motion of a segment during the intervals when it is bearing an input.

π^S bounds: servo capability. During the free window $W_{i,k}$, no input is on segment k . Friction does not bind, because there is no input to slip. The kinematic bounds that apply to π^S are therefore the servo's own bounds: $a_{\max}^S$ and $j_{\max}^S$, which are set by the motor and drive train rather than by contact friction. These are strictly greater than the π^I bounds — typically by a substantial margin, because friction is the dominant constraint in the in-contact regime.

This bound separation is the reason the per-segment apparatus has explanatory power. If π^I and π^S shared the same bounds, the buffer would be a single coupled control problem; the segmentation would be cosmetic. The fact that π^S enjoys a higher kinematic budget than π^I is what makes the free window an exploitable resource and what makes the feasibility test $W_{i,k} \geq T_{\min}(\zeta_{i,k})$ a meaningful gate rather than a trivial one.

Jerk boundary conditions are not enforced. The boundary state ξ at each crossing event records velocity and acceleration but not jerk. The model choice is that π^I is piecewise linear in jerk — admitting infinite snap at segment boundaries — and π^S inherits this convention for

consistency. Jerk continuity across the boundary is not required, and the physical justification is that the snap discontinuity is absorbed by the elasticity of the drive train without observable consequence at the scale of the input motion. This is the same modeling choice made throughout the controls literature on piecewise-constant-jerk trajectories.

2.5 Operational Manifestation

In the simulator, the per-segment quantities of this section manifest as the `SegmentEvents` and `SegmentSyncResponse` types in `ksb.analysis`. `SegmentEvents` carries $t_{i,k}^{\text{in}}$, $t_{i,k}^{\text{out}}$, $\xi_{i,k}^{\text{in}}$, $\xi_{i,k}^{\text{out}}$, and $W_{i,k}$ as arrays indexed by pair and segment. `SegmentSyncResponse` derives $T_{\min}(\zeta_{i,k})$ via a per-cell Ruckig call under π^S bounds, and exposes the kinematic margin $W_{i,k} - T_{\min}(\zeta_{i,k})$ and feasibility test $W_{i,k} \geq T_{\min}(\zeta_{i,k})$. Both the document and the code are 0-indexed in pairs and segments, so the correspondence is direct.

3. The Two Readings of the Margin

The kinematic margin is a time quantity: $M_{i,k} = W_{i,k} - T_{\min}(\zeta_{i,k})$ measures, in seconds, the surplus or deficit of the free window against the time the transition demands. This is the **time reading**, and it is the natural object for failure-mode characterization — it is policy-free, computable from $\zeta_{i,k}$ and $W_{i,k}$ alone, and exactly the quantity the simulator records. A negative time margin says the segment runs out of time; the three regimes of §2.2 are read directly off its sign and the sign of $W_{i,k}$.

There is a second, equivalent reading. Instead of asking how much time the segment lacks, ask what kinematic state it actually delivers at the receiving boundary when it runs out. A segment that cannot complete its transition arrives at $t_{i+1,k}^{\text{in}}$ not at the demanded $\xi_{i+1,k}^{\text{in}}$ but at some other state, short by a velocity and an acceleration: a residual pair $(\Delta v, \Delta a)$. This is the **state reading** — the synchronization deficit expressed in the space of kinematic states rather than in time.

The two readings carry the same information. A feasible transition has zero time deficit and zero state error simultaneously; an infeasible one has both. They cross zero together, and either can serve as the feasibility test. The difference is what each makes convenient. The time reading is policy-free and is therefore the right diagnostic for asking *whether and where* the buffer fails — it needs no commitment to what the segment does when it cannot finish. The state reading is not policy-free: a residual $(\Delta v, \Delta a)$ exists only once a degradation policy is fixed, because the segment must do *something* specific when it runs out of time — saturate jerk and carry a velocity residual, or truncate the profile and carry an acceleration residual — and the split between Δv and Δa depends on that choice. But this is exactly what makes the state reading the right object for a different question: not whether the buffer fails, but *what the failure physically is* and how it propagates into the next segment's boundary condition. The time reading localizes the failure; the state reading describes it.

The rest of this work uses the time reading throughout, because its questions are feasibility questions — where, how often, under what load and variability does $M_{i,k}$ go negative. The state reading is the bridge to a control-theoretic treatment of the buffer, where the residual at one boundary becomes the disturbance the next segment must reject, and it is the natural language for any future work that takes up the degradation policy rather than the feasibility boundary.